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| **Fiscal Year:** | FY 2011 | **Task Last Updated:** | FY 02/04/2011 |
| **PI Name:** | Stutte, Gary Ph.D. | | |
| **Project Title:** | Mutualism in a Reduced Gravity Environment (MuRGE): Pirasporoforma indica: Arabidopsis thaliana Interactions in Microgravity | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology<br>(3) Plant Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Immunology | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | NASA CENTER | **Phone:** | 321-861-3493 |
| **Organization Name:** | QinetiQ-North America/NASA Kennedy Space Center | | |
| **PI Address 1:** | ESC-Team QNA | | |
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| **City:** | Kennedy Space Center | **State:** | FL |
| **Zip Code:** | 32899 | **Congressional District:** | 24 |
| **Comments:** | NOTE: PI had a change of employment from Dynamac Corporation to QinetiQ-North America, effective 1 March, 2011. | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2009 Space Life Sciences BRIC NNH09ZTT004N |
| **Start Date:** | 03/19/2010 | **End Date:** | 10/18/2010 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA KSC |
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| **Flight Program:** | Shuttle/ISS | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Roberts, Michael ( Dynamac ) | | |
| **Grant/Contract No.:** | NNX10AG93G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | MuRGE (Mutualism in a Reduced Gravity Environment) will investigate microgravity effects associated with cell-cell communication and beneficial microbe-host interactions using a plant-fungal model system. Seeds from the model plant species Arabidopsis thaliana will be germinated and grown on-orbit in the presence or absence of Piriformospora indica, an endophytic fungus of the Sebacinaceae family. P. indica is a capable of colonizing the roots of a wide variety of plant species, including non-mycorrhizal hosts like A. thaliana, and promoting plant growth in a manner similar to arbuscular mycorrhizal fungi (AMF). Unlike most AMF, P. indica is not an obligate plant symbiont and can be grown from spores or hyphae in axenic culture in the absence of a host. In the presence of a suitable plant host, P. indica can attach to and colonize root tips. The interaction can be visualized by the appearance of strong auto-fluorescence in the roots, followed by colonization of root cells by fungal hyphae and production of chlamydospores. Increased root growth can be observed even before root colonization is detectable. This proposal will use cellular and morphological analyses of fixed A. thaliana specimens and P. indica cultivated in microgravity separately in axenic culture or together in co-culture for comparison to ground controls to determine the effect of microgravity on the formation of a mutualistic endosymbiosis between plant and fungus. In addition, P. indica chlamydospores generated from axenic culture in microgravity will be used to inoculate roots of A. thaliana grown in 1g to determine the effect of microgravity upon the inherent virulence or beneficial effect of P. indica in association with a plant host. Based on recent reports of increased virulence of S. typhimurium, P. aeruginosa, and S. pneumoniae in reduced gravity, differences in microbial pathogenic responses and host plant systemic acquired resistance are expected. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The mutualism that develops between Piriformospora indica and a plant system results in greater growth of the host plant, enhanced resistance to disease, and potential improvements of nutritional quality. These initial experiments established that a mutualism can be established between P. indica and Arabidopsis thaliana, thus validating a model system to understand the fundamental mechanism regulating these responses. The understanding of the mechanisms that result in mutualism is important, since there are many parallels to fungal pathogens in the infection and colonization of the host plant. However, in these special cases, the response actually enhances the growth of the host. Identification of the biochemical or biomolecular cues that prevent the infection from becoming pathogenic is the ultimate goal of future work with this model system. |
| **Task Progress:** | Experiments were conducted to establish the feasibility of investigating plant/fungal interactions in microgravity using the Arabidopsis thaliana: Piriformospora indica model system. Key findings of the experiments are:<br>• P. indica will produce chlamydospores in the culture without a plant host.<br>• A. thaliana planting densities up to 18 seeds/60 mm Petri dish can be achieved.<br>• P. indica: A. thaliana mutualism develops, plant growth increases and spore formation results when conducted in relevant spaceflight environments.<br>• Protocols for preflight production and quantification of P. indica were developed.<br>• Protocols for post-flight analysis of A. thaliana and P. indica growth and development were validated.<br>• Benchmarks for P. indica: A. thaliana mutualism growth responses were established.<br>It was concluded that MuRGE science objectives are feasible as outlined in NRA proposal entitled “Mutualism in a Reduced Gravity Environment (MuRGE): Piriformospora indica: Arabidopsis thaliana Interactions in Microgravity,” submitted in response to NASA Research Announcement NNH09ZTT004N: Research Opportunities for Flight Experiments in Space Life Sciences: Biological Research in Canisters for Arabidopsis thaliana. |
| **Bibliography Type:** | Description: (Last Updated: 04/16/2020) |